

| Question | Marking Guidance                                                                                                                                                                                                                                                                             | Mark  | Comments                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07.1     | <p>1. (Antivenom/Passive immunity) antibodies bind to the toxin/venom/antigen and (causes) its destruction;</p> <p>2. Active immunity would be too slow/slower;</p>                                                                                                                          | 2     | <p>1. For 'bind' accept 'attach', ignore 'attack'.</p> <p>1. For 'destruction of toxin' accept agglutination or phagocytosis.</p> <p>1. Ignore reference to antibodies 'neutralising toxin/stopping damage'</p> <p>1. Reject reference to 'killing' toxin/venom.</p> <p>2. Accept 'passive immunity is <u>faster</u>', not simply 'passive immunity is fast'.</p> |
| 07.2     | <p>1. May be different form of antigen/toxin (within one species)</p> <p>OR</p> <p>Snakes (within one species) may have different mutations/alleles;</p> <p>2. Different antibodies (needed in the antivenom)</p> <p>OR</p> <p>(Several) antibodies complementary (to several antigens);</p> | 2 max | No mark points are available for answers related to collecting venom from different species of snake.                                                                                                                                                                                                                                                             |
| 07.3     | <p>1. Horses <b>because</b> more antivenom/antibodies could be collected (as more blood collected);</p> <p>2. 4550 (cm<sup>3</sup>) v 26 (cm<sup>3</sup>) (blood collected);</p>                                                                                                             | 2     | <p>2. Accept 175 rabbits needed to (collect the volume of blood from) one horse.</p>                                                                                                                                                                                                                                                                              |

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|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07.4     | <ol style="list-style-type: none"> <li>(So) the animal does not suffer from the venom/vaccine/toxin;</li> <li>(So) the animal does not suffer anaemia/does not suffer as a result of blood collection;</li> <li>(So) the animal does not have pathogen that could be transferred to humans;</li> </ol>                                                            | 1 max | <p>Accept 'To fulfil licence/legal requirements'.</p> <p>Accept '(So) the animal does not have pathogen that could result in it producing other antibodies (not wanted in the antivenom)'.</p> <p>For 'pathogen' accept correct form of pathogen.</p>                                                                                                                                                                   |
| 07.5     | <ol style="list-style-type: none"> <li>B cells specific to the venom reproduce by mitosis;</li> <li>(B cells produce) plasma cells and memory cells;</li> <li>The second dose produces antibodies (in secondary immune response) in higher concentration <b>and</b> quickly</li> </ol> <p>OR</p> <p>The first dose must be small so the animal is not killed;</p> | 3     | <ol style="list-style-type: none"> <li>Accept in context of primary or secondary immune response.</li> <li>Credit idea of specificity if given once in relation to T or B cell.</li> <li>Accept a description for specificity.</li> <li>Accept 'clone' for 'reproduce by mitosis'.</li> <li>'Clonal selection of B cells' = MP1.</li> <li>Accept 'a lot of antibody' for 'higher concentration of antibody'.</li> </ol> |

